ONLINE SHOPPING AND MULTI-VENDOR E-COMMERCE SYSTEM

CAPSTONE PROJECT

ACTIVITY DOCUMENT

SUBMITTED BY

SHYAM M

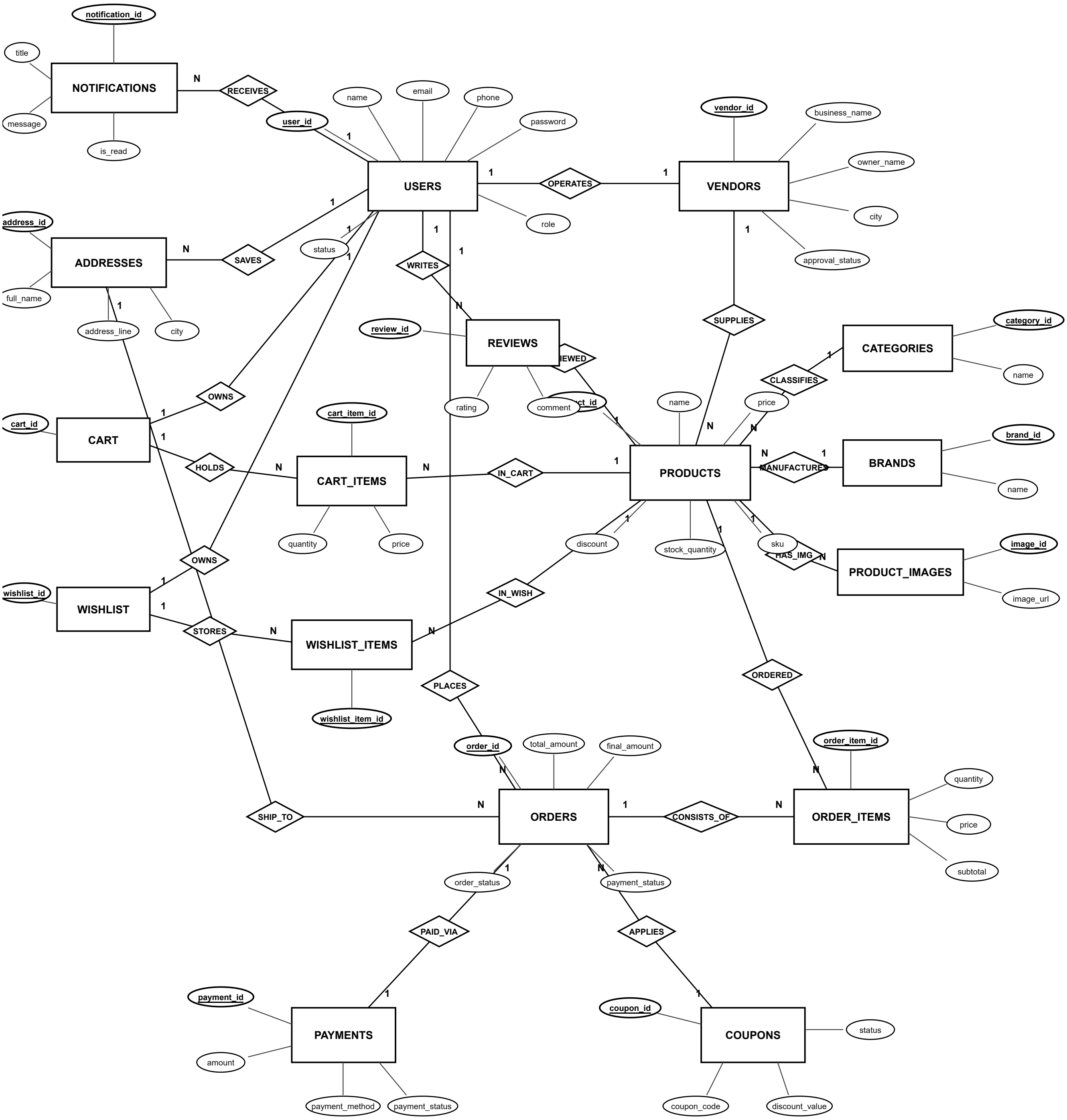
YEAR: III

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

J.J COLLEGE OF ENGINEERING AND TECHNOLOGY

INDEX PAGE

TRADITIONAL ER DIAGRAM



TRADITIONAL ER DIAGRAM EXPLANATION

The Traditional Entity Relationship (ER) Diagram represents the overall database structure of the Online Shopping and Multi-Vendor E-Commerce System (BuyIt Marketplace). It illustrates the major entities, their attributes, relationships, and cardinalities used in the system. The ER Diagram helps in understanding how different entities are connected before converting the design into relational database tables.

The USERS entity is one of the main entities in the system. It stores information such as user ID, name, email, phone number, password, role, status, and account creation details. Users can act as customers, vendors, or system administrators depending on their assigned role. A customer can place multiple orders, maintain a personal cart and wishlist, save delivery addresses, receive notifications, and write product reviews.

The VENDORS entity stores business and merchant information for registered sellers on the marketplace. Each vendor record is associated with a user through the user_id in a one-to-one relationship. The entity contains business name, owner name, business description, address, city, state, pincode, and administrative approval status. A vendor can list and supply multiple products in the catalog.

The CATEGORIES entity organizes marketplace products into logical departments such as Electronics, Mobiles, Fashion, Home, and Books. It includes category name, description, image URL, and status. Each category can contain multiple products, establishing a one-to-many relationship with the products entity.

The BRANDS entity stores information about manufacturers and brands associated with marketplace items. It includes brand name, description, brand logo URL, and active status. Multiple products can belong to a specific brand.

The PRODUCTS entity stores important product catalog details such as product name, description, price, discount percentage, stock quantity, SKU code, display image, and availability status. Products form the central part of the e-commerce system. Each product belongs to a specific vendor, category, and brand, and can have multiple gallery images, reviews, and order items.

The PRODUCT_IMAGES entity stores multiple images related to products. A product can have multiple images, allowing the system to display different views and detailed gallery photos of the item to prospective buyers.

The ADDRESSES entity stores delivery address information created by customers. Each address record contains details such as recipient full name, phone number, address line, city, state, postal pincode, and default address flag. A user can store multiple addresses for shipping.

The CART entity manages active shopping sessions for customers. Each customer has one cart record connected via customer_id. The CART_ITEMS entity represents individual products placed in the cart, capturing the desired quantity and item price. Every cart item belongs to a specific cart and references a product.

The WISHLIST entity enables customers to save products of interest for future purchase. Each customer is associated with one wishlist through customer_id. The WISHLIST_ITEMS entity maintains the junction between the wishlist and individual products, allowing users to bookmark multiple items.

The ORDERS entity is responsible for managing customer checkout transactions and order fulfillment. It connects the USERS entity (customer) with the ADDRESSES entity (shipping destination). Order records maintain total amount, shipping charges, discount deductions, final payable amount, payment status, and order status. A user can place multiple orders over time.

The ORDER_ITEMS entity records the individual product line items contained within each order. It captures an immutable historical snapshot of the purchased item, including product ID, vendor ID, product name at the time of purchase, unit price, quantity, and line-item subtotal. Each order item connects to a specific order and product.

The PAYMENTS entity stores payment transaction records associated with customer orders. It includes details such as payment amount, payment method (Cash on Delivery, UPI, Card, or Net Banking), payment date, payment status, and transaction reference ID. Payments are associated with orders in a one-to-one relationship to maintain rigorous financial accounting records.

The COUPONS entity stores promotional discount information such as unique coupon code, discount type (percentage or fixed amount), discount value, minimum order requirement, maximum discount limit, usage limits, and validity dates. Coupons can be applied during checkout to provide promotional savings.

The REVIEWS entity stores feedback and ratings provided by customers for products they have purchased. It includes rating score (1 to 5 stars), review comments, moderation status, and creation date. A user can write reviews for multiple products, helping prospective buyers evaluate item quality.

The NOTIFICATIONS entity stores system alerts and operational messages sent to users. A user can receive multiple notifications related to order confirmations, shipping updates, payment receipts, or promotional offers. Notification details include message title, body text, type, read status, and timestamp.

Overall, the Traditional ER Diagram provides a clear representation of the entities and relationships used in the Online Shopping and Multi-Vendor E-Commerce System. Primary Keys are used to uniquely identify records, while Foreign Keys establish relationships between related entities. The ER Diagram was later converted into relational tables using schema mapping rules, ensuring data integrity, consistency, and proper database organization.

SCHEMA MAPPING RULE

1. Entity to Table Mapping

Each entity in the ER Diagram was converted into a separate relational table. For example, USERS, VENDORS, CATEGORIES, BRANDS, PRODUCTS, ORDERS, and PAYMENTS were converted into database tables.

2. Attribute to Column Mapping

Each attribute of an entity was converted into a column in the corresponding table, retaining standard descriptive identifiers such as user_id, price, stock_quantity, and order_status.

3. Primary Key Mapping

Each table was assigned a Primary Key to uniquely identify every record, implemented using auto-incrementing SERIAL integer sequences.

4. Relationship Mapping

Relationships between entities were implemented using Foreign Keys. For example, the customer_id in the ORDERS table references the USERS table, and vendor_id in the PRODUCTS table references the VENDORS table.

5. One-to-Many Relationship Mapping

For a one-to-many relationship, the Primary Key of the parent table was added as a Foreign Key in the child table. For example, category_id is added to PRODUCTS to reference the parent CATEGORIES table.

6. One-to-One Relationship Mapping

For a one-to-one relationship, a Foreign Key with a UNIQUE constraint was used. For example, user_id in the VENDORS table and customer_id in the CART table uniquely reference the USERS table.

7. Many-to-Many Relationship Mapping

Many-to-many relationships were represented using an intermediate or junction table when required. For example, the relationship between CART and PRODUCTS is represented via CART_ITEMS, and between ORDERS and PRODUCTS via ORDER_ITEMS, avoiding data redundancy.

8. Foreign Key Mapping

Foreign Keys were used to establish relationships between related tables. For example, customer_id connects USERS with ORDERS, ADDRESSES, CART, WISHLIST, REVIEWS, and NOTIFICATIONS.

9. Referential Integrity

Referential integrity was maintained by using Foreign Key constraints with appropriate cascading actions (ON DELETE CASCADE, SET NULL, or RESTRICT). This ensures that a record in a child table cannot reference a non-existing record in a parent table.

10. Constraint Mapping

Constraints such as NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK (e.g. price >= 0, stock_quantity >= 0, rating between 1 and 5), and DEFAULT were applied to maintain data accuracy, consistency, and integrity.

11. Data Type Mapping

Appropriate data types were selected based on the type of information stored: VARCHAR for text values, INT for numerical values, DATE and TIMESTAMP for date-related information, and DECIMAL(10,2) for exact monetary and currency values.

DATA DICTIONARY

The following data dictionary describes the tables, columns, data types, keys, and constraints used in the Online Shopping and Multi-Vendor E-Commerce System database.

USERS Table

Column Name	Data Type	Key / Constraint
user_id	SERIAL	PRIMARY KEY
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	UNIQUE, NOT NULL
phone	VARCHAR(20)	-
password	VARCHAR(255)	NOT NULL
role	VARCHAR(20)	DEFAULT 'CUSTOMER'
status	VARCHAR(20)	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

VENDORS Table

Column Name	Data Type	Key / Constraint
vendor_id	SERIAL	PRIMARY KEY
user_id	INT	NOT NULL, UNIQUE, FOREIGN KEY
business_name	VARCHAR(200)	NOT NULL
owner_name	VARCHAR(100)	NOT NULL
description	TEXT	-
address	TEXT	-
city	VARCHAR(100)	-
state	VARCHAR(100)	-
pincode	VARCHAR(10)	-
approval_status	VARCHAR(20)	DEFAULT 'PENDING'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

CATEGORIES Table

Column Name	Data Type	Key / Constraint
category_id	SERIAL	PRIMARY KEY
name	VARCHAR(100)	NOT NULL
description	TEXT	-
image	VARCHAR(500)	-
status	VARCHAR(20)	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

BRANDS Table

Column Name	Data Type	Key / Constraint
brand_id	SERIAL	PRIMARY KEY
name	VARCHAR(100)	NOT NULL
description	TEXT	-
logo	VARCHAR(500)	-
status	VARCHAR(20)	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

PRODUCTS Table

Column Name	Data Type	Key / Constraint
product_id	SERIAL	PRIMARY KEY
vendor_id	INT	NOT NULL, FOREIGN KEY
category_id	INT	FOREIGN KEY
brand_id	INT	FOREIGN KEY
name	VARCHAR(255)	NOT NULL
description	TEXT	-
price	DECIMAL(10,2)	NOT NULL
discount	DECIMAL(5,2)	DEFAULT 0
stock_quantity	INT	NOT NULL, DEFAULT 0
sku	VARCHAR(50)	-
image	VARCHAR(500)	-
status	VARCHAR(20)	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

PRODUCT_IMAGES Table

Column Name	Data Type	Key / Constraint
image_id	SERIAL	PRIMARY KEY
product_id	INT	NOT NULL, FOREIGN KEY
image_url	VARCHAR(500)	NOT NULL

ADDRESSES Table

Column Name	Data Type	Key / Constraint
address_id	SERIAL	PRIMARY KEY
customer_id	INT	NOT NULL, FOREIGN KEY
full_name	VARCHAR(100)	NOT NULL
phone	VARCHAR(20)	-
address_line	TEXT	NOT NULL
city	VARCHAR(100)	NOT NULL
state	VARCHAR(100)	NOT NULL
pincode	VARCHAR(10)	NOT NULL
is_default	BOOLEAN	DEFAULT FALSE

CART Table

Column Name	Data Type	Key / Constraint
cart_id	SERIAL	PRIMARY KEY
customer_id	INT	NOT NULL, UNIQUE, FOREIGN KEY
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

CART_ITEMS Table

Column Name	Data Type	Key / Constraint
cart_item_id	SERIAL	PRIMARY KEY
cart_id	INT	NOT NULL, FOREIGN KEY
product_id	INT	NOT NULL, FOREIGN KEY
quantity	INT	NOT NULL, DEFAULT 1
price	DECIMAL(10,2)	NOT NULL

WISHLIST Table

Column Name	Data Type	Key / Constraint
wishlist_id	SERIAL	PRIMARY KEY
customer_id	INT	NOT NULL, UNIQUE, FOREIGN KEY
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

WISHLIST_ITEMS Table

Column Name	Data Type	Key / Constraint
wishlist_item_id	SERIAL	PRIMARY KEY
wishlist_id	INT	NOT NULL, FOREIGN KEY
product_id	INT	NOT NULL, FOREIGN KEY
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

ORDERS Table

Column Name	Data Type	Key / Constraint
order_id	SERIAL	PRIMARY KEY
customer_id	INT	NOT NULL, FOREIGN KEY
address_id	INT	FOREIGN KEY
total_amount	DECIMAL(10,2)	NOT NULL, DEFAULT 0.00
shipping_amount	DECIMAL(10,2)	DEFAULT 0.00
discount_amount	DECIMAL(10,2)	DEFAULT 0.00
final_amount	DECIMAL(10,2)	NOT NULL, DEFAULT 0.00
payment_status	VARCHAR(20)	DEFAULT 'PENDING'
order_status	VARCHAR(30)	DEFAULT 'PLACED'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

ORDER_ITEMS Table

Column Name	Data Type	Key / Constraint
order_item_id	SERIAL	PRIMARY KEY
order_id	INT	NOT NULL, FOREIGN KEY
product_id	INT	NOT NULL, FOREIGN KEY
vendor_id	INT	NOT NULL, FOREIGN KEY
product_name	VARCHAR(255)	NOT NULL
price	DECIMAL(10,2)	NOT NULL
quantity	INT	NOT NULL
subtotal	DECIMAL(10,2)	NOT NULL
item_status	VARCHAR(30)	DEFAULT 'PLACED'

PAYMENTS Table

Column Name	Data Type	Key / Constraint
payment_id	SERIAL	PRIMARY KEY
order_id	INT	NOT NULL, FOREIGN KEY
payment_method	VARCHAR(20)	NOT NULL
transaction_id	VARCHAR(100)	-
amount	DECIMAL(10,2)	NOT NULL
payment_status	VARCHAR(20)	DEFAULT 'PENDING'
payment_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

COUPONS Table

Column Name	Data Type	Key / Constraint
coupon_id	SERIAL	PRIMARY KEY
coupon_code	VARCHAR(50)	UNIQUE, NOT NULL
discount_type	VARCHAR(20)	NOT NULL
discount_value	DECIMAL(10,2)	NOT NULL
minimum_amount	DECIMAL(10,2)	DEFAULT 0
maximum_discount	DECIMAL(10,2)	-
start_date	TIMESTAMP	-
expiry_date	TIMESTAMP	-
usage_limit	INT	DEFAULT 0
used_count	INT	DEFAULT 0
status	VARCHAR(20)	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

REVIEWS Table

Column Name	Data Type	Key / Constraint
review_id	SERIAL	PRIMARY KEY
product_id	INT	NOT NULL, FOREIGN KEY
customer_id	INT	NOT NULL, FOREIGN KEY
rating	INT	NOT NULL
comment	TEXT	-
status	VARCHAR(20)	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

NOTIFICATIONS Table

Column Name	Data Type	Key / Constraint
notification_id	SERIAL	PRIMARY KEY
user_id	INT	NOT NULL, FOREIGN KEY
title	VARCHAR(200)	NOT NULL
message	TEXT	NOT NULL
type	VARCHAR(50)	-
is_read	BOOLEAN	DEFAULT FALSE
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP